Library Management Analysis SQL project

Our Database Schema

```sql
create database librarydataanalysis;

use librarydataanalysis;

-- Drop existing tables if they exist
DROP TABLE IF EXISTS borrowings;
DROP TABLE IF EXISTS books;
DROP TABLE IF EXISTS members;
```

Creating Books table

```sql
-- Creating table named books
create table books (
id SERIAL PRIMARY KEY,
  title TEXT NOT NULL,
  author TEXT NOT NULL,
  category TEXT NOT NULL
);
```

Creating Borrowings table

```sql
CREATE TABLE borrowings (
  id SERIAL PRIMARY KEY,
  book_id INT REFERENCES books(id),
  member_id INT REFERENCES members(id),
  borrow_date DATE NOT NULL,
  return_date DATE
);
```

Creating Members table

```sql
CREATE TABLE members (
  id SERIAL PRIMARY KEY,
  name TEXT NOT NULL,
  join_date DATE NOT NULL
);
```

Populating sample data into books table:

```sql
INSERT INTO books (title, author, category) VALUES
('The Hobbit', 'J.R.R. Tolkien', 'Fantasy'),
('The Fellowship of the Ring', 'J.R.R. Tolkien', 'Fantasy'),
('The Two Towers', 'J.R.R. Tolkien', 'Fantasy'),
('The Return of the King', 'J.R.R. Tolkien', 'Fantasy'),
('To Kill a Mockingbird', 'Harper Lee', 'Fiction'),
('1984', 'George Orwell', 'Fiction'),
('Pride and Prejudice', 'Jane Austen', 'Romance'),
('The Great Gatsby', 'F. Scott Fitzgerald', 'Fiction'),
('Clean Code', 'Robert C. Martin', 'Programming'),
('Design Patterns', 'Erich Gamma', 'Programming'),
("Man Search for Meaning","Viktor Frankl","Psychology"),
("Wolf Worm", 'T. Kingfisher', "Horror"),
("Japanese Gothic", "Kylie Lee Baker", "Horror"),
("Not Quite Dead Yet", "Holly Jackson", "Crime Thriller"),
("The Tenant", "Freida McFadden", "Crime Thriller"),
("Beautiful Ugly", "Alice Feeney", "Crime Thriller");
```

Insert data into members table:

```sql
INSERT INTO members (`name`, join_date) VALUES
('Katherine', '2024-05-03'),
('Luis', '2023-09-27'),
('Oliver', '2024-07-12'),
('Maya', '2023-04-16'),
('Ranvijay', '2024-05-16'),
('Charlie', '2024-03-05'),
('Diana', '2024-03-20'),
('Ethan', '2024-04-01'),
('Navpreet', '2024-09-08'),
('Hiren', '2023-01-19'),
('Ashish', '2024-06-25');
```

Inserting data into the borrowings table (some returned, some overdue):

```sql
INSERT INTO borrowings (book_id, member_id, borrow_date, return_date) VALUES
(1, 1, '2025-01-01', '2025-01-20'),
(14, 9, '2025-01-14',NULL),    -- overdue if > 30 days
(2, 2, '2025-02-01', NULL),
(5, 2, '2025-01-15', '2025-01-25'),
(16, 2, '2025-02-20', NULL),          -- overdue
(4, 8, '2025-02-18', '2025-02-23'),
(6, 3, '2025-02-10', '2025-02-20'),
(10, 3, '2025-03-01', NULL),         -- recently borrowed
(7, 9, '2025-03-03', NULL),
(11, 4, '2025-02-05', '2025-02-28'),
(15, 5, '2025-02-07', '2025-02-12'),
(9, 4, '2025-03-10', NULL),
(12, 5, '2025-03-01', NULL),          -- still with member
(13, 5, '2025-03-05', NULL);         -- still with member
```

Data Highlights:

Books – A mix of different genres including Fiction, fantasy, romance, programming, psychology and crime thriller.

Members – 11 people (some joined in 2023 while some in 2024)

Borrowings --

Some books returned on time.

Some overdue (borrowed > 30 days ago, still not returned).

Some recently borrowed.

Some members borrowed multiple books in the same month.

Let's look at our Tables:

Books table -

id	title	author	category
1	The Hobbit	J.R.R. Tolkien	Fantasy
2	The Fellowship of the Ring	J.R.R. Tolkien	Fantasy
3	The Two Towers	J.R.R. Tolkien	Fantasy
4	The Return of the King	J.R.R. Tolkien	Fantasy
5	To Kill a Mockingbird	Harper Lee	Fiction
6	1984	George Orwell	Fiction
7	Pride and Prejudice	Jane Austen	Romance
8	The Great Gatsby	F. Scott Fitzgerald	Fiction
9	Clean Code	Robert C. Martin	Programming
10	Design Patterns	Erich Gamma	Programming
11	Man Search for Meaning	Viktor Frankl	Psychology
12	Wolf Worm	T. Kingfisher	Horror
13	Japanese Gothic	Kylie Lee Baker	Horror
14	Not Quite Dead Yet	Holly Jackson	Crime Thriller
15	The Tenant	Freida McFadden	Crime Thriller
16	Beautiful Ugly	Alice Feeney	Crime Thriller

Members table –

id	name	join_date
1	Katherine	2024-05-03
2	Luis	2023-09-27
3	Oliver	2024-07-12
4	Maya	2023-04-16
5	Ranvijay	2024-05-16
6	Charlie	2024-03-05
7	Diana	2024-03-20
8	Ethan	2024-04-01
9	Navpreet	2024-09-08
10	Hiren	2023-01-19
11	Ashish	2024-06-25

Borrowings table –

id	book_id	member_id	borrow_date	return_date
1	1	1	2025-01-01	2025-01-20
2	14	9	2025-01-14	NULL
3	2	2	2025-02-01	NULL
4	5	2	2025-01-15	2025-01-25
5	16	2	2025-02-20	NULL
6	4	8	2025-02-18	2025-02-23
7	6	3	2025-02-10	2025-02-20
8	10	3	2025-03-01	NULL
9	7	9	2025-03-03	NULL
10	11	4	2025-02-05	2025-02-28
11	15	5	2025-02-07	2025-02-12
12	9	4	2025-03-10	NULL
13	12	5	2025-03-01	NULL
14	13	5	2025-03-05	NULL

Solving some common Business Problems:

Q:) Display some most borrowed books-

```sql
SELECT
  b.id,
  b.title,
  b.author,
  COUNT(br.id) AS borrow_count
FROM books b
LEFT JOIN borrowings br ON b.id = br.book_id
GROUP BY b.id, b.title, b.author
ORDER BY borrow_count DESC, b.title
LIMIT 5;
```

Output:

id	title	author	borrow_count
6	1984	George Orwell	1
16	Beautiful Ugly	Alice Feeney	1
9	Clean Code	Robert C. Martin	1
10	Design Patterns	Erich Gamma	1
13	Japanese Gothic	Kylie Lee Baker	1

Q:) List the library members with some overdue books (borrowed >30 days and not returned)

```sql
SELECT
  m.id,
  m.name,
  br.book_id,
  b.title,
  br.borrow_date
FROM members m
JOIN borrowings br ON m.id = br.member_id
JOIN books b ON br.book_id = b.id
WHERE br.return_date IS NULL
  AND br.borrow_date < (CURRENT_DATE - INTERVAL 30 DAY )
ORDER BY br.borrow_date;
```

Output:

id	name	book_id	title	borrow_date
9	Navpreet	14	Not Quite Dead Yet	2025-01-14
2	Luis	2	The Fellowship of the Ring	2025-02-01
2	Luis	16	Beautiful Ugly	2025-02-20
3	Oliver	10	Design Patterns	2025-03-01
5	Ranvijay	12	Wolf Worm	2025-03-01
9	Navpreet	7	Pride and Prejudice	2025-03-03
5	Ranvijay	13	Japanese Gothic	2025-03-05
4	Maya	9	Clean Code	2025-03-10

Q:) Display Category wise most popular book:

```sql
WITH book_counts AS (
  SELECT
    b.id,
    b.title,
    b.category,
    COUNT(br.id) AS borrow_count
  FROM books b
  LEFT JOIN borrowings br ON b.id = br.book_id
  GROUP BY b.id, b.title, b.category
),
```

```sql
category_max AS (
  SELECT
    category,
    MAX(borrow_count) AS max_count
  FROM book_counts
  GROUP BY category
)
```

```sql
  SELECT
    bc.category,
    bc.id   AS book_id,
    bc.title,
    bc.borrow_count
  FROM book_counts bc
  JOIN category_max cm
    ON bc.category = cm.category
   AND bc.borrow_count = cm.max_count
  ORDER BY bc.category;
```

Output:

category	book_id	title	borrow_count
Crime Thriller	14	Not Quite Dead Yet	1
Crime Thriller	15	The Tenant	1
Crime Thriller	16	Beautiful Ugly	1
Fantasy	1	The Hobbit	1
Fantasy	2	The Fellowship of the Ring	1
Fantasy	4	The Return of the King	1
Fiction	5	To Kill a Mockingbird	1
Fiction	6	1984	1
Horror	12	Wolf Worm	1
Horror	13	Japanese Gothic	1
Programming	9	Clean Code	1
Programming	10	Design Patterns	1
Psychology	11	Man Search for Meaning	1
Romance	7	Pride and Prejudice	1

Q:) Display members who borrowed more than 1 books in the same month:

```sql
WITH member_month_counts AS (
  SELECT
    br.member_id,
    DATE_FORMAT(br.borrow_date, '%Y-%m-01') AS month_start,
    COUNT(br.id) AS books_borrowed
  FROM borrowings br
  GROUP BY br.member_id, DATE_FORMAT(br.borrow_date, '%Y-%m-01')
)
```

```sql
SELECT
  mm.member_id,
  m.name,
  mm.month_start,
  mm.books_borrowed
FROM member_month_counts mm
JOIN members m ON mm.member_id = m.id
WHERE mm.books_borrowed > 1
ORDER BY mm.member_id, mm.month_start;
```

Output:

member_id	name	month_start	books_borrowed
2	Luis	2025-02-01	2
5	Ranvijay	2025-03-01	2